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"It all depends on the end user and what he wants to do with the kit. For education purpose, the main aspects are the openness of the system, price of the kit and the ecosystem available with it. For research, what's important is the flexibility the kit gives to change the specifications of the robot for various applications, product development or R&D using off-the-shelf components," says Fahad Azad, managing partner, Robosoft Systems India.

Deshpande opines, "An ideal hobby kit would be a microcontroller development board with schematics, standard sensors, reliable wireless modules, direct-current motors with encoders and a USB interface. The development board really helps to interface many different kinds of peripherals that can be developed on your own."

"If you plan to use the kit for outdoor purpose, make sure that it is hermetically sealed to be dust- and water-proof. For rough terrains or climbing purpose, the robot should ideally use tracks and not wheels," Col. N.C. Pande (Retd), joint director-training, EFY TechCenter.

"Also, check whether the kit comes with a well-documented manual or a training program. Make sure that the kit is expandable so that you can keep developing more and more complex projects," he adds.

Specifications play a very important role. So make sure you have the right specifications.

Dr. Kavi Arya, associate professor, CSE department, IIT-Bombay, highlights, "In a robotic kit for students, ruggedness, capability of interfacing with a wide variety of hardware, good software support in terms of language and tool and community support are important points to consider. For instance, check whether it can be programmed using C (are integrated development environments available), has been ported to programming environments such as that provided by Microsoft's Robotic Studio, or is compatible with Open Source environments such as Player/Stage?"

Quality of the kit is also a major concern. "Quality applies to all the components, beginning with the motors. Many companies sell gear motors with plastic gear box, which basically fails to perform. Also, a lot of Chinese stuff is available in the market," says Amar Gala, partner, Mac-Net Technology.

"Go for a kit that is modular, re-configurable and reusable for different applications. Going for low cost would necessitate compromising on the quality or functionality for reducing the initial expense. The additional expense can be recovered over multiple use of the kit," suggests Jayakrishnan.

"If cost is the most critical factor, it is better to go for raw spare parts and build it from scratch for a limited functionality. This at least will help one to ensure quality at a lower cost," he adds.

"A good robotic kit should grow with the user's knowledge and experience. In the stage of basic experiments, you should be able to perform simple experiments like line followers and obstacle avoiders. In intermediate stage, you should be able to add more advanced microcontrollers and communication modules. In advanced stage, you should be able to add hardware platforms that support real-time operating systems over which you can have image processing, path planning, etc," suggests Dev.

Most kits come with a warranty. High-valued kits generally provide full warranty against manufacturing defects and service warranty against wear and tear over usage. In case of software, one-year free upgrade can be considered as a standard policy.

"In a nutshell, the various factors to keep in mind are high quality of mechanical elements, customisable processing elements, expandability through open interfaces, expandable design, i.e., ability to reuse design from simple to complex, supported multiple communication protocols, low power requirements, ease of mechanical construction and easy chaining of processing elements to increase the computing power," Sudhanshu sums up. ●

The author is an assistant editor at EFY

with advanced environments like C and C++," opines Sudhanshu.

How to select a kit?

As explained earlier, the foremost thing is to define the application of the kit.